Open-charm extension of AMPT v1.26t9b/v2.26t9b: what was changed, where, and why

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Abstract

This note lists every change made to the public AMPT model v1.26t9b/v2.26t9b [1, 2] to study open charm in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. For each change it gives the physics motivation and the paper it follows, the file, routine and source lines that implement it, the switch that turns it on, and the check that it works. All changes are either a patch to the seven stock source files (170 lines added, 17 changed) or four new files (`charm_mod.f`, `coal_new.f`, `hqctl.inc`, `eps09s_f.cpp`). Every change is behind an environment switch that defaults to off, so with no switch set the program behaves as stock AMPT. The appendix tables are generated from the repository (`docs/tools/code_map.py`), so the line numbers are those of the patched tree that `scripts/build.sh` creates in `build/ampt-src/`.

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1 How to inspect the changes yourself

1. `scripts/setup_deps.sh` downloads the official AMPT archive from Z.-W. Lin's web page and checks its SHA-256 (`9441b5f7...`). The seven Fortran files of that archive are the base; no other version is used.

2. `scripts/build.sh` unpacks it, applies `patches/ampt-v1.26t9b-v2.26t9b_charm.patch` and copies the four new files of `src_new/`. Read the patch directly: it has 46 hunks (Table 1), each opening with a comment that starts with `cdmeson`. To list them: `grep -n cdmeson build/ampt-src/*.f`.

3. With no `AMPT_*` variable set, every new routine returns at once and every stock branch is taken. This was checked byte for byte on `ana/ampt.dat` against the unmodified program (gate G0 of the development log). `configs/stock.env` runs the program this way.

4. `tests/test_exact_reproduction.sh` regenerates events of the paper production from their seeds and compares them with the files written on the batch farm (Sec. 6).

2 Event flow and where the charm code enters

The AMPT string-melting chain [1] is unchanged in structure: HIJING [3] (initial conditions, strings) → string melting → ZPC parton cascade [4] → quark coalescence → ART hadron cascade [5]. The framework adds hooks at each stage:

stage	what is added	routine (file)
initialisation	read all switches	`HQINIT` (`charm_mod.f`), called from `main.f`
HIJING hard process	triggered $c\bar{c}$ pair, modern PDFs, nPDF weight, HIJING trigger bug fix	`hijing1.383_ampt.f`, `hipyset1.35.f`; `HQXPDF`, `HQNMOD`, `HQSHWT`, `HQTRGG`
strings → partons	charm momenta taken before Lund fragmentation	`HQSAVE`, `HQPUT`, `HQHTOP`
ZPC	charm cross section, isotropic charm scattering, collision counters, local freeze-out fixes	`zpc.f`, `HQFAC`, `HQCOLL`, `HQFRZG`
coalescence	He–Lin coalescence with charm r_{BM}; charm coalescence criterion and Peterson fragmentation; $D^*/D = 1$	`coal_new.f`; `HQFRAG`, `HQDSTR` (called from `linana.f`)
ART	optional charm-hadron rescattering (not used in the reference configuration)	`HQHADR` (called from `art1f.f`)
output	`ana/charm.dat` with charm records at every stage	`HQHTOP`, `HQOUTF`, `HQOUTH`

3 The changes

Each subsection gives: **why** (motivation and reference), **what** (the algorithm), **where** (routine and lines; Tables 1–3 give all line numbers), **switch** and its value in the reference configuration, and **check**.

3.1 Heavy-quark trigger and a HIJING bug fix

Why. Charm is rare: about one $c\bar{c}$ pair per hundred minimum-bias p+p events at RHIC. Simulating charm observables with 10^6 untriggered Au+Au events would give too few D mesons. HIJING

has a built-in trigger (`IHPR2(3)=3`) that forces one hard $c\bar{c}$ pair into each event [3]. The improved heavy-flavour AMPT of Zheng *et al.* [6] uses it without a transverse-momentum cut on the pair.
What. `AMPT_HQTRIG=1` sets `IHPR2(3)=3` and `HIPR1(10)=AMPT_HQPTMIN` (default 0). One primary nucleon–nucleon collision, chosen at random, gets a $q\bar{q}, gg \to c\bar{c}$ hard scattering (PYTHIA `MSUB` 81/82). The number of primary NN collisions of the event (HIJING `NCOLT`) is written to the event record, because the analysis weights each triggered event by it (Sec. 5).
Bug fix. `JETINI` (`hijing1.383_ampt.f`) initialises PYTHIA lazily per collision type. Its first-initialisation branch returned without updating the remembered type `ilast`. The next untriggered minijet call then skipped the reset of `MSUB`/`CKIN`, so every later minijet of the event was generated with the trigger process list: test events carried 24–92 charm quarks instead of about 2. The fix sets `ilast=itrig` before that `RETURN` (tag `cdmeson-fix`). It is active only when the trigger is on; public AMPT does not use the trigger.
Where. `HQINIT`; `hijing1.383_ampt.f` hunks at the NN-collision count and `JETINI` (Table 1).
Reference: `AMPT_HQTRIG=1`, `AMPT_HQPTMIN` unset (0).
Check. Net charm is zero in every event (analysis counter `netc_nonzero_events`=0 on 2.4M events).

3.2 One clean pair per event

Why. With the trigger, untriggered minijets and parton showers can still produce extra charm. That breaks the one-pair-per-event bookkeeping on which the event weight rests.
What. `AMPT_HQCLEAN=1` removes `MSUB` 81/82 from the untriggered minijets and sets `MSTJ(45)=3` (no $g \to c\bar{c}$ in final-state showers). **Where.** `HQINIT`; `hijing1.383_ampt.f` (`AMPT_HQCLEAN` hunk).
Reference: on.

3.3 Modern proton PDFs for the triggered pair

Why. HIJING 1.383 uses the Duke–Owens PDFs at leading order. With them the $c\bar{c}$ cross section at 200 GeV is 6.6 times below STAR's, and the D-meson spectrum shape is distorted. Zhang *et al.* [7] and Zheng *et al.* [6] replaced them with CTEQ6.1M in AMPT.
What. With `AMPT_HQPDF`≥ 1, `PYSTFU` returns CTEQ6.1M [13] (via LHAPDF 6 [14], set `cteq61`, member 0) for the *triggered hard process only*; minijets and the bulk keep HIJING's PDFs. The factorisation scale is $\mu_F^2 =$`AMPT_HQMUF`$\times Q^2$ (default 2, i.e. $2m_T^2$ as in Ref. [6]). PYTHIA's maximum search is raised by `AMPT_HQXSF` (default 1.5) so that the acceptance–rejection step stays valid with the new, larger PDFs.
Where. `HQXPDF` (`charm_mod.f`); hooks in `hipyset1.35.f` (`PYSTFU`, `PYSIGH`, `PYMAXI`) and `hijing1.383_ampt.f` (`JETINI`). **Reference:** `AMPT_HQPDF=1`.
Check. The p+p D^0 spectrum shape against STAR gives $\chi^2/\mathrm{ndf} = 3.5/5$, against 6.7/5 for public AMPT (`observation/results/tables/PP_SHAPE.txt`).

3.4 Impact-parameter-dependent nuclear PDFs (EPS09s) as an event weight

Why. This is the change behind the main result of the paper. The charm pair is produced at a point in the transverse plane, and its nuclear modification depends on the transverse position of the colliding nucleons in their nuclei. Position-dependent shadowing therefore changes *where* charm is produced relative to the bulk, and hence the charm rapidity-odd "tilt" that drives D^0 directed flow. EPS09s [15] is the global nPDF fit with a spatial dependence; we use the authors' code and tables unmodified.
What. The pair is sampled with free-nucleon CTEQ6.1M PDFs. Its nuclear modification $w = R_1(x_1, Q, s_1)\, R_2(x_2, Q, s_2)$ is then carried as an event weight (`AMPT_HQSHW=1`). R is the EPS09s NLO bound-to-free ratio of the incoming parton flavour, with isospin conjugation for neutrons, at the transverse distance s of each nucleon from its own nucleus centre (Ref. [15], Eq. 2.9). The weight

is written to field 18 of the `charm.dat` event record and applied in the analysis. Sampling plus weighting is equivalent to sampling with nuclear PDFs (gate g4 of the development log), keeps the events identical between nPDF variants, and lets the 31 EPS09s error sets be applied offline to the same events.
Where. `HQNMOD`, `HQSHWT`, `HQTRGG` (`charm_mod.f`); `eps09s_f.cpp` (C bindings to the official `eps09s.cpp`); hooks in `PYSTFU/PYRAND` (`hipyset1.35.f`). **Reference:** `AMPT_HQSHW=1` with `AMPT_HQPDF=1`. A related option, `AMPT_HQSHAD=s`, scales HIJING's own shadowing of the triggered process ($R \to 1 + s(R-1)$). It was used only for the shadowing scan of the paper.

3.5 Charm momentum taken before string fragmentation (direct charm)

Why. In string melting, charm quarks are produced from hadrons made by Lund fragmentation, which softens them. Zhang *et al.* [8, 9] and Fan *et al.* [10] take the charm quarks directly from the HIJING hard scattering. Their spectrum is much harder.
What. `AMPT_HQDIRECT=1`. Before `HIJFRG`, `HQSAVE` stores every $c/\bar{c}$ on each string. After string melting, `HQHTOP` gives each melted charm quark the momentum of the closest saved charm parton of the same flavour on the same string, with formation time $0.197\,E/m_T^2$ fm. The 3-momentum difference goes to the light partons of that string, energy-weighted, so string 3-momentum is conserved exactly. The event's energy change (a few GeV out of ~ 40 TeV) and the matching counters are written to the event record.
Where. `HQSAVE`, `HQPUT`, `HQHTOP`; hooks in `hijing1.383_ampt.f` and `linana.f`. **Reference:** on.
Check. Matching failures `nfail`=0 in every event of the production.

3.6 Charm–parton cross section and isotropic charm scattering in ZPC

Why. With the stock forward-peaked cross section, charm couples too weakly to the medium: D^0 $v_2 \approx 0.04$ against 0.15 in data. The improved AMPT uses separate light- and heavy-quark cross sections [9].
What. `AMPT_SIGHQ=r` multiplies the closest-approach criterion $d^2 < \sigma/\pi$ by r for every pair containing a c or $\bar{c}$, in all twelve `isco*` routines (`HQFAC`). `AMPT_HQISO=1` makes charm collisions isotropic in the pair c.m. frame (same random deviate, so the event sequence is preserved), for pair c.m. momenta below `AMPT_HQISO_PCUT`.
Reference. The input screening mass is 1.6016 fm^{-1}, i.e. $\sigma_{LQ} = 6$ mb, and `AMPT_SIGHQ=1.5`, i.e. $\sigma_{HQ} = 9$ mb, isotropic below 0.5 GeV. These values were chosen jointly on D^0 v_2 and R_{AA}.

3.7 Local parton freeze-out

Why. The reference input uses AMPT's `ISOFT=5` option ("local freeze-out" in Lin's `linana.f`): a parton freezes when its local energy density falls below a critical value, rather than after its last collision. In the stock code, a parton that never satisfies the criterion before the cascade ends is hadronised with its *initial* momentum. We fix that.
What. `AMPT_ECRIT` sets the critical energy density (stock 1.0 $\mathrm{GeV/fm}^3$). `HQFRZG` freezes a never-frozen parton at its last interaction, as `ISOFT=4` does (0.07 % of partons are affected). With `AMPT_HQLATE=0`, charm quarks never freeze by the density criterion. They keep scattering, changing only their own momentum, until the cascade ends, and are then frozen at their last interaction.
Where. `amptsub.f` (`ARTSET`), `HQFRZG`, `zpc.f`, `linana.f` `LOCAL`. **Reference:** `ISOFT=5`, `AMPT_ECRIT=1.0`, `AMPT_HQLATE=0`.

3.8 Hadronisation: improved coalescence, charm coalescence criterion and fragmentation

Why. The public AMPT coalescence forms mesons and baryons from fixed quark groupings and builds antibaryons from antiquarks a median 13 fm apart. He and Lin's improved coalescence [11] chooses the closest partners and lets a parameter r_{BM} decide between baryon and meson formation; it is published but not distributed, so we reimplemented it. For charm, Zheng *et al.* [6] use a larger r_{BM} to produce enough Λ_c. Zhang *et al.* [9] and Fan *et al.* [10] add a coalescence criterion: charm hadrons whose constituents are too far apart, too massive or at too high m_T fragment instead (Peterson fragmentation [12]).
What.

- `AMPT_COALNEW=1`: He–Lin coalescence (`coal_new.f`) with $r_{BM} = 0.61$ for light flavours and r^c_{BM} =`AMPT_RBMHQ` for groups containing charm. `AMPT_CNORDER=2` sweeps the partons in a random order per event, removing a projectile/target bias of the stored order.

- `AMPT_HQFRAG=1` (`HQFRAG`): an open-charm candidate coalesces if $m_T < m_T^{\max}$ (`AMPT_HQMTMAX`), or if its largest constituent separation is below p_r (`AMPT_HQPR`) and $m_{\rm inv} < \sum m_q + p_m(m_H - \sum m_q)$ (`AMPT_HQPM`), with constituent quark masses (`AMPT_HQCONST=1`, Fan *et al.* Eq. 2) (`AMPT_HQMTMODE=1`: the "or" of Ref. [10]). Otherwise the charm quark fragments. z is drawn from the Peterson function ($\epsilon_c = 0.05$), the light antiquark comes from a vacuum pair ($u : d : s = 1 : 1 : 0.3$), and the remaining momentum goes to a light hadron, so 3-momentum is conserved per candidate. The hadron record carries a mode flag (1 coalescence, 2 fragmentation). `AMPT_HQBDIST=1` (systematic variant c1) uses the Jacobi distance $\sqrt{2/3}\,|r_c - (r_{q_1} + r_{q_2})/2|$ for baryons.

- `AMPT_DSTAR1=1` (`HQDSTR`): primordial $D^*/D = 1$ (and $D^*_s/D_s = 1$), as in Ref. [6]. The charm–light candidates are ordered by excess mass; the lower half become D, the upper half D^*.

Reference: `AMPT_COALNEW=1`, `AMPT_RBMHQ=3`, `AMPT_CNORDER=2`, `AMPT_HQFRAG=1`, $p_r = 0.5$ fm, $p_m = 0.5$, $m_T^{\max} = 3.3$ (Au+Au) and 2.0 GeV (p+p), `AMPT_HQCONST=1`, `AMPT_DSTAR1=1`.

3.9 Output

`AMPT_CHARM_OUT=1` writes `ana/charm.dat` with one record type per stage. E: event, with b, the participants, $N_{\rm coll}$, the energy change, the triggered $\sigma_{c\bar{c}}$ and the nPDF weight. I/L: charm and a light-parton sample at formation. F/G: the same at freeze-out. H: charm hadrons after coalescence, with mode and constituent separation. Z: the end of the parton phase. A private random generator is used, so writing the file never changes the events. `AMPT_LQFRAC` sets the light-parton sampling fraction. The node-side reduction `scripts/reduce_dmeson_v2.awk` keeps every charm hadron, all charged kaons and protons, a 25 % (10 % forward) sample of pions with weights, and the full second-order flow vectors of all charged hadrons in the STAR and PHENIX event-plane windows.

3.10 Options implemented but not used in the reference configuration

- `AMPT_HQHAD=1` (`HQHADR`): elastic, isotropic charm–hadron rescattering in ART with the He–Fries–Rapp cross sections (7 mb with mesons, 10 mb with baryons) [16]. It was tested on the reference configuration: 0.17 collisions per charm hadron, v_2 unchanged, R_{AA} worse by 4–14 %. Not adopted; quoted as a systematic in the paper.

- `AMPT_HQCRONIN=`δ (`HQCKICK`): Cronin k_T broadening of the pair, Zhang *et al.* [8] Eqs. 1–3 (used by Fan *et al.*; a candidate improvement for R_{AA}).

- `AMPT_ZPC_TSTOP`: hard stop of the parton cascade, Zhang and Xu [17].
- `AMPT_TILT`: a sensitivity knob (light partons shifted by $\lambda\eta_s$), not a physics model.
- `AMPT_SOUT`, `AMPT_HQDREST`, `AMPT_HQFBAR`, `AMPT_HQPDFSET`, `AMPT_HQCDBG`: diagnostics and variants.

4 Reference configuration

Input card `inputs/auau200_lf_s6.ampt` = the stock `input.ampt` with `ISOFT=5` and screening mass 1.6016 fm^{-1} (the event count, impact parameter range and seeds are set per job). Switches: `configs/reference_auau.env` and `configs/reference_pp.env` (Table 3). Public AMPT for comparison: `inputs/auau200_sm.ampt` (the stock card) with only the trigger and the output switches (`configs/public_*.env`).

arm	config	events	HIJING seed base	ZPC se
dm24_e_a, dm24_e_b	reference_auau	$2 \times 1000 \times 300$	20261107, 20261108	$13i + 9$
dm25_e_c…h	reference_auau	$6 \times 1000 \times 300$	20261109…14	$31i + 1$
dm24_epp, dm25_epp	reference_pp	$2 \times 200 \times 30000$	20261107, 20261115	$13i + 9$
dm25_c1 / dm25_ppc1	c1_auau / c1_pp	1000×300 / 100×30000	20261116 / 20261117	$31i + 1$
dm25_pub_a, b / dm25_pppub	public_*	$2 \times 1000 \times 300$ / 100×30000	20261118, 19 / 20261120	$31i + 2$

HIJING seed of job i = base $+ 7919\,i$. Au+Au: $0 < b < 14.5$ fm; p+p: $0 < b < 2$ fm.

5 Analysis conventions

- **Feed-down.** $D^{*0} \to D^0$, $D^{*+} \to D^0\pi^+$ (67.7 %), $D_s^* \to D_s$, $\Sigma_c \to \Lambda_c$ (PDG branching ratios, `analysis/dmlib.py`), as in the measured yields.
- **Weights.** Each triggered Au+Au event carries one pair, while real charm production scales with $N_{\rm coll}$. Charm observables are therefore weighted by the event's `NCOLT` times the EPS09s weight; the charm-quark count by `NCOLT` only. Light hadrons are never weighted.
- R_{AA} = (D^0 per c quark, Au+Au) / (D^0 per c quark, p+p). This equals the usual definition when the total charm yield scales with $N_{\rm coll}$ times the nPDF factor.
- **Centrality.** Charged-particle multiplicity in $|\eta| < 0.5$, as STAR.
- **Flow.** v_2: the scalar product in STAR's η-sub-event geometry, with the resolution taken per 5 % bin; v_1: relative to the reaction plane (STAR uses the spectator plane), with the sign fixed by HIJING's projectile at $x = +b/2$.
- **Errors.** Jackknife over job files (delete-a-block, at most 2000 blocks).

6 Verification

- **Stock reproduction (G0).** With no switch set, `ana/ampt.dat` is byte-identical to the unmodified program.
- **Exact reproduction of the production** (`make test-repro`). The first events of Au+Au job 1 and p+p job 1 of the production, regenerated on a different machine and compiler (gfortran 13.3 vs 11.5), give a byte-identical `charm.dat` and identical event headers, flow vectors and charm hadrons.

- **Independent re-analysis** (`make test-quick`). `analysis/indep_check.py` re-implements the observables from the documentation without importing the production analysis. On 36k Au+Au and 300k p+p events it agrees with `analysis/analyze_arm.py` to 10^{-17}.
- **Results** (`make test-results`). Every table and figure in `observation/results/` is regenerated from the shipped analysis outputs.

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A Code map (generated from the repository)

Table 1: Every hunk of `patches/ampt-v1.26t9b-v2.26t9b_charm.patch`. Lines = the added lines in the patched file `build/ampt-src/`; +/− = lines added/removed. The note is the first comment of the hunk (tagged `cdmeson` in the source).

file	lines	+/−	note
`amptsub.f`	44	+1/−0	code: include 'hqctl.inc'
`amptsub.f`	52–53	+2/−0	cdmeson (src_v9): AMPT_ECRIT overrides it (read by HQINIT)
`amptsub.f`	4923–4925	+3/−1	cdmeson (src_v9): partons never frozen keep their final, not initial,
`art1f.f`	619–621	+3/−1	dmeson src_v17: charm-hadron rescattering (returns at once unless

file	lines	+/−	note
hijing1.383_ampt.f	199–200	+2/−0	cdmeson src_v13: ihqtrx (trigger context of AMPT_HQSHAD)
hijing1.383_ampt.f	583–599	+12/−0	cdmeson: number of primary NN collisions of this event (E record)
hijing1.383_ampt.f	648–655	+6/−0	cdmeson src_v14: with AMPT_HQPDF the triggered PYINIT (maximum search)
hijing1.383_ampt.f	1346–1349	+3/−0	cdmeson: record charm partons on the strings before Lund fragmentation
hijing1.383_ampt.f	4298–4299	+2/−0	cdmeson:
hijing1.383_ampt.f	4396–4400	+5/−0	cdmeson: AMPT_HQCLEAN, no heavy-quark pair production in untriggered jets
hijing1.383_ampt.f	4527–4547	+18/−1	cdmeson src_v14: AMPT_HQPDF raises every triggered maximum by hqxsf:
hipyset1.35.f	7603–7604	+2/−0	cdmeson src_v14:
hipyset1.35.f	7874–7875	+2/−0	cdmeson src_v14: largest weight/maximum inside the triggered process
hipyset1.35.f	7970–7971	+2/−0	cdmeson src_v15: AMPT_HQSHW=1, nuclear modification as event weight
hipyset1.35.f	12017–12018	+2/−0	cdmeson src_v14:
hipyset1.35.f	12144–12147	+3/−0	cdmeson src_v14: hard-process call (AMPT_HQMUF scales muF$\hat{\}$2 here only)
hipyset1.35.f	13981–13982	+2/−0	cdmeson src_v13:
hipyset1.35.f	14331–14336	+6/−0	cdmeson src_v14: AMPT_HQPDF, triggered hard process: CTEQ6.1M (x EPS09s
hipyset1.35.f	14575–14578	+4/−0	cdmeson src_v14: AMPT_HQPDF already applied EPS09s (no HIJING shadowing)
hipyset1.35.f	14597–14598	+2/−1	cdmeson src_v13: AMPT_HQSHAD scales the triggered hard process only
linana.f	656–657	+2/−0	cdmeson: direct-charm option, ZPC charm counters, charm.dat records
linana.f	734–738	+5/−1	cdmeson: charm partons at ZPC freeze-out
linana.f	929–930	+2/−0	cdmeson: AMPT_DSTAR1, primordial D*/D = 1 (Zheng et al. 2020)
linana.f	1099–1102	+4/−0	cdmeson: AMPT_HQFRAG, coalescence criterion + Peterson fragmentation
linana.f	1998	+1/−0	code: include 'hqctl.inc'
linana.f	2047–2057	+11/−2	cdmeson src_v12b (AMPT_HQLATE=x>=0): a charm quark freezes only below
main.f	179–180	+2/−1	cdmeson: D-meson framework switches (charm_mod.f), all default off
zpc.f	1121	+1/−0	code: include 'hqctl.inc'
zpc.f	1134–1140	+7/−0	cdmeson (src_v9): AMPT_ZPC_TSTOP, hard stop of the partonic cascade at
zpc.f	1864–1865	+2/−0	cdmeson: count charm scatterings
zpc.f	1972–2001	+24/−3	cdmeson src_v12 (AMPT_HQLATE>=0): an unfrozen charm quark still scatters
zpc.f	2196–2207	+12/−0	cdmeson src_v12: the frozen partner of a late charm collision is untouched
zpc.f	2223	+1/−0	code: logical hqisof
zpc.f	2239–2240	+2/−0	cdmeson: AMPT_HQISO, isotropic scattering for charm (same deviate rx)
zpc.f	3936	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4064	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)

file	lines	+/−	note
zpc.f	4199	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4349	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4505	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4669	+1/−1	code: allok = allok .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4802	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	4966	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	5146	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	5294	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	5462	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)
zpc.f	5646	+1/−1	code: allokp = allokp .and. dm2 .lt. cutof2*hqfac(i, j)

Table 2: Routines of the new files (`src_new/`), with their line ranges and the first line of their header comment.

file	routine	lines	purpose
charm_mod.f	HQBLK	72–92	
charm_mod.f	HQINIT	93–258	read the switches; called from main.f before HIJSET
charm_mod.f	HQFAC	259–273	ZPC: factor on cutof2 (=sigma/pi) for the pair (i,j) in /prec2/
charm_mod.f	HQCOLL	274–306	ZPC: count 2-body scatterings of charm partons (/prec2/ index)
charm_mod.f	HQFRZG	307–348	ISOFT=5, end of ZPC (zpstrg, before /frzprc/ replaces /prec2/):
charm_mod.f	HQSAVE	349–398	HIJING, string-melting branch. ISG=0: reset for this event and
charm_mod.f	HQPUT	399–419	
charm_mod.f	HQHTOP	420–592	end of HTOP: (a) AMPT_HQDIRECT replacement, (b) reset ZPC charm
charm_mod.f	hqrand	593–605	private LCG for output subsampling; never touches RANART/NSEED so
charm_mod.f	HQOUTF	606–646	start of PTOH (ZPC finished): charm partons at freeze-out (F) with
charm_mod.f	HQOUTH	647–715	end of PTOH: charm hadrons right after coalescence (H) with the
charm_mod.f	HQNCOL	716–723	HIJING: store the number of primary NN collisions of the event
charm_mod.f	HQDSTR	724–827	AMPT_DSTAR1 (PTOH, right after the flavour/species loop): set the
charm_mod.f	HQFRAG	828–1156	end of PTOH (hadron momenta/positions set): coalescence criterion
charm_mod.f	KBARY	1157–1186	lightest ground-state baryon (KF > 0) of three light flavours
charm_mod.f	KCBARY	1187–1205	ground-state singly-charmed baryon (KF > 0) c + light diquark
charm_mod.f	HQXPDF	1206–1248	PYSTFU, triggered hard process only: x*f of a proton from CTEQ6.1M
charm_mod.f	HQNMOD	1249–1299	src_v15 (moved out of HQXPDF, same arithmetic): EPS09s NLO bound-
charm_mod.f	HQSHWT	1300–1353	src_v15 AMPT_HQSHW=1, PYRAND after the channel choice of the
charm_mod.f	HQRRX	1354–1383	src_v15: HIJING 1.383 nuclear shadowing factor exactly as PYSTFU
charm_mod.f	hqrnp	1384–1391	EPS09s power series 1 + c1 T + c2 T$\hat{}${}2 + c3 T$\hat{}${}3 + c4 T$\hat{}${}4
charm_mod.f	HQTRGG	1392–1409	HIJING, at the choice of the triggered NN collision: number of
charm_mod.f	HQCKICK	1410–1569	AMPT_HQCRONIN=delta, end of HTOP after the HQDIRECT replacement:
charm_mod.f	HQLBST	1570–1585	Lorentz boost of the four-vector q (px,py,pz,E) by velocity b

file	routine	lines	purpose
charm_mod.f	HQVMAX	1586–1602	once per event: report a new maximum of SIGS/XSEC_max inside the
charm_mod.f	HQHADR	1603–1815	src_v17 AMPT_HQHAD=1: hadronic rescattering of charm hadrons in
charm_mod.f	HQBST4	1816–1835	src_v17: Lorentz boost of the 4-vector x = (x1,x2,x3,x0) into the
charm_mod.f	hqrnh	1836–1846	src_v17: private Park-Miller generator of the charm rescattering
coal_new.f	CNPAR	1–72	He & Lin's improved quark coalescence, PRC 96 014910 (arXiv:1703.02673),
coal_new.f	CNDIS2	73–115	Relative distance between two flat-list partons in AMPT's own convention:
coal_new.f	CNDIS3	116–163	He & Lin's d_B: the average of the three relative distances among three
coal_new.f	COALNEW	164–366	Replaces 'coales' when AMPT_COALNEW=1.
coal_new.f	CNCHK	367–462	Gate 1: parton bookkeeping, as an assertion. A hadronisation rewrite that
coal_new.f	CNWRIT	463–527	Write the chosen groups back into the SOFT arrays and reset NSG, so that

Table 3: Environment switches: the source line that reads each one, and its value in the reference Au+Au and p+p configurations (`configs/reference_*.env`). Unset = the stock behaviour.

switch	read at	reference Au+Au	reference p+p
AMPT_HQTRIG	charm_mod.f:100	1	1
AMPT_HQCLEAN	charm_mod.f:102	1	1
AMPT_HQDIRECT	charm_mod.f:104	1	1
AMPT_CHARM_OUT	charm_mod.f:106	1	1
AMPT_DSTAR1	charm_mod.f:108	1	1
AMPT_HQISO	charm_mod.f:110	1	unset
AMPT_HQISO_PCUT	charm_mod.f:112	0.5	unset
AMPT_TILT	charm_mod.f:114	unset	unset
AMPT_ECRIT	charm_mod.f:119	1.0	1.0
AMPT_SOUT	charm_mod.f:124	unset	unset
AMPT_HQLATE	charm_mod.f:126	0	0
AMPT_HQSHAD	charm_mod.f:132	unset	unset
AMPT_HQPDF	charm_mod.f:138	1	1
AMPT_HQPDFSET	charm_mod.f:140	unset	unset
AMPT_HQMUF	charm_mod.f:142	unset	unset
AMPT_HQXSF	charm_mod.f:144	unset	unset
AMPT_HQDREST	charm_mod.f:158	unset	unset
AMPT_HQSHW	charm_mod.f:163	1	unset
AMPT_HQCONST	charm_mod.f:168	1	1
AMPT_HQBDIST	charm_mod.f:174	unset	unset
AMPT_HQHAD	charm_mod.f:177	unset	unset
AMPT_HQHADM	charm_mod.f:179	unset	unset
AMPT_HQHADB	charm_mod.f:181	unset	unset
AMPT_HQHADDS	charm_mod.f:183	unset	unset
AMPT_HQHADLC	charm_mod.f:185	unset	unset
AMPT_HQHADREC	charm_mod.f:187	unset	unset
AMPT_HQCRONIN	charm_mod.f:197	unset	unset
AMPT_ZPC_TSTOP	charm_mod.f:202	unset	unset
AMPT_HQMTMODE	charm_mod.f:207	1	1
AMPT_HQFBAR	charm_mod.f:209	unset	unset
AMPT_HQFRAG	charm_mod.f:211	1	1

switch	read at	reference Au+Au	reference p+p
AMPT_HQPR	charm_mod.f:213	0.5	0.5
AMPT_HQPM	charm_mod.f:215	0.5	0.5
AMPT_HQMTMAX	charm_mod.f:217	3.3	2.0
AMPT_HQEPS	charm_mod.f:219	unset	unset
AMPT_HQPTMIN	charm_mod.f:221	unset	unset
AMPT_SIGHQ	charm_mod.f:223	1.5	1.5
AMPT_LQFRAC	charm_mod.f:225	0.05	unset
AMPT_HQCDBG	charm_mod.f:879	unset	unset
AMPT_COALNEW	coal_new.f:24	1	1
AMPT_CNORDER	coal_new.f:40	2	2
AMPT_RBM	coal_new.f:47	unset	unset
AMPT_RBMHQ	coal_new.f:57	3	3